Adaptive Threat Detection for Securing Cloud CI/CD Pipelines Using Evolving Neural Architectures

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AUTHORS

Shifat Ahmed (22301582)
Sahil Ishan Tonmoy (22301612)
S.M. Mashrafi Zaman (22301614)
Shafayet Karim (22101507)

SUPERVISOR

Mr. Partha Bhoumik
Lecturer
Department of Computer Science and Engineering
Brac University

01. Abstract

Cloud-based Continuous Integration and Continuous Deployment (CI/CD) pipelines enable rapid software delivery but are increasingly vulnerable to advanced cyberattacks such as polymorphic and metamorphic malware. Traditional security solutions rely on fixed rules and known signatures, limiting their ability to detect evolving threats. This research proposes an adaptive threat detection framework integrated directly into cloud CI/CD pipelines. The system combines source code analysis and runtime behavior monitoring using self-evolving deep learning models. A threat mutation module generates controlled malware variants, enabling continuous learning without full retraining. By dynamically adapting to new attack patterns, the approach improves detection accuracy, reduces false positives.

02. Reasearch Objective

1. Design an architecture for a self-adaptive neural network.

2. Integrate Threat Mutation for Adversarial Learning.

3. Enhance Pipeline Security through Continuous Self-Learning.

4. Establish CI/CD Pipelines with Real-Time Monitoring.

5. Evaluate and Benchmark the Adaptive Detection Framework.

03. Literature Review

Research Papers	*Key Strategy*	*Feature*
Kalash et al. [2]	*M-CNN on malware images (Malimg)*	*Converts binaries to grayscale images; CNN extracts visual patterns for classification.*
Kolosnjaji et al. [1]	*ConvNet-LSTM on system-call Sequences*	*Dynamic, sequence-based malware classification from sandboxed*
Catak et al. [4]	*LSTM on API call sequences (Mal-API-2019)*	*Sequential deep learning on malware samples; effective for temporal*
Oliveira and Sass [3]	*Graph CNN on API call sequences*	*Converts API sequences to graphs; GCN learns behavioral patterns*

04. Dataset

Training data were derived from the Malware Analysis Datasets [21], containing benign and malicious samples represented as grayscale PE images and dynamic API call sequences. This enables the model to learn complementary spatial structure and temporal behavioral patterns directly from the raw data.



Figure: Dataset samples

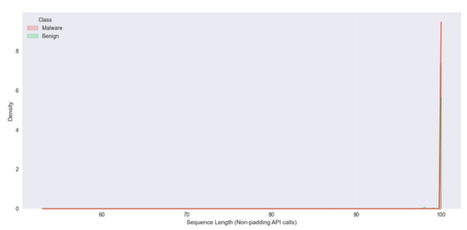


Figure: Sequence length distribution

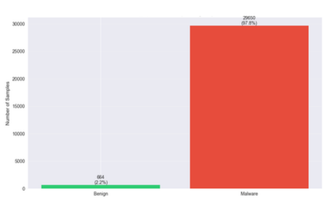
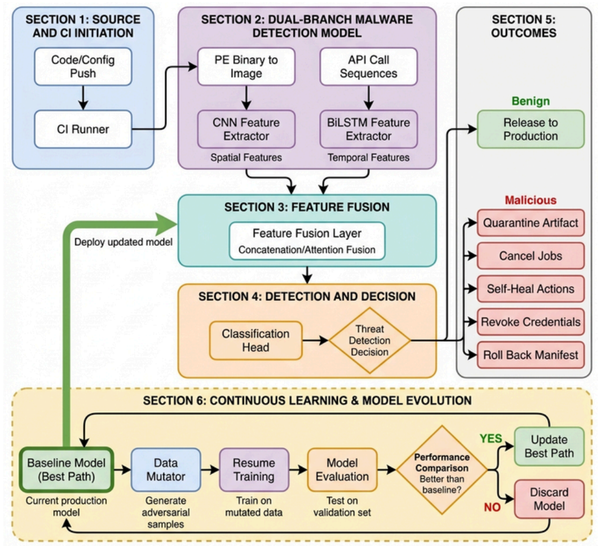


Figure: Class distribution

05. Methodology

An end-to-end adaptive framework integrating dual-branch deep learning with automated CI/CD pipelines to continuously detect and evolve against mutating malware.



06. Feature Fusion Comparison

Method	Recall (Malware)	Specificity (Benign)	Accuracy	Verdict
Concat (Baseline)	98.38%	92.25%	98.25%	Best Recall & Acc
Add	97.81%	90.85%	97.66%	Worst
Attention	97.83%	93.66%	97.74%	Best Specificity

Figure: Classification comparsion between fusion strategies

Concatenation proves to be the superior fusion strategy (98.25% accuracy) by preserving full feature context, outperforming the specialized precision of Attention and the lossy compression of Addition.

07. Mutatuion Strategy Comparsion

Strategy	Clean Acc.	Insert (30%)	Replace (30%)	Append (30%)
Random Insert	96.8%	95.4%	93.4%	96.8%
Random Replace	90.8%	70.8%	69.8%	90.8%
Noise Append	94.0%	92.0%	86.0%	94.0%

Figure: Robustness comparsion between mutation strategies

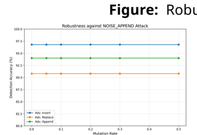


Figure: Robustness against Noise append attack

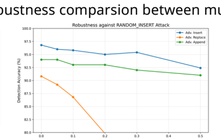


Figure: Robustness against Random insert attack



Figure: Robustness against Random replace attack

08. Analysis



Figure: Confusion Matrix

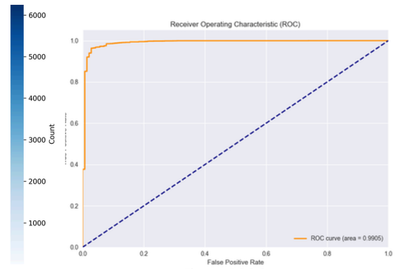


Figure: ROC Curve

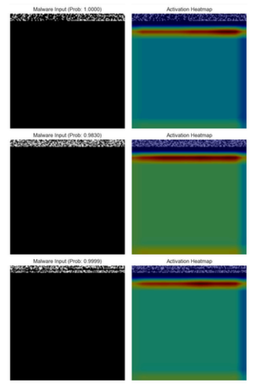


Figure: Activation heatmap

Class	Precision	Recall	F1-Score	Support
Benign	0.560	0.922	0.697	142
Malware	0.998	0.984	0.991	6354
Accuracy			**0.983**	**6496**

Figure: Classification report

09. Result

The comparative results show that the proposed dual-branch model (CNN + BiLSTM) achieves the highest accuracy, precision, and recall among all referenced methods. By jointly capturing spatial and sequential features, our system outperforms single-architecture and graph-based approaches. These results demonstrate the effectiveness of the proposed model for robust and accurate malware detection.

Reference	Method	Accuracy	Precision	Recall
Our System	**Dual-Branch (CNN + BiLSTM)**	**98.65%**	**99.76%**	**98.87%**
Kalash et al. (2018) *[2]*	M-CNN (Deep CNN on Images)	98.52%	98.50%	98.50%
Kolosnjaji et al. (2016) *[1]*	CNN + LSTM (Sequential API)	97.70%	98.10%	97.50%
Catak et al. (2020) *[4]*	Standard LSTM (API Sequence)	95.00%	95.00%	95.00%
Oliveira & Sassi (2019) *[3]*	Graph CNN (API Graphs)	92.44%	~92.00%	~92.00%

Figure: Comparison table against other sytems

10. Recommendations for Future Work

- **Expanded Dataset Diversity:** Extend the dual-branch architecture to support additional file formats (e.g., PDF) and integrate larger, real-world datasets to improve generalization beyond the Kaggle corpus.
- **Generative Adversarial Networks (GANs):** Replace the current random mutation logic with GAN-based adversarial training to generate sophisticated, realistic evasion samples, hardening the model against state-of-the-art polymorphic attacks.

11. Reference

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